TAEYOUNG YUN

Ph.D student @ KAIST

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RESEARCH INTEREST

My research interests lie in controllable generative modeling with large models (e.g., LLMs, Diffusion / Flow-based models) by exploring their latent spaces. In particular, I'm interested in building an amortized sampler that can extract crucial latents to generate desired samples. To accomplish this, my research focuses on amortized inference for generative models using off-policy reinforcement learning (RL) methods.

I've also participated in several transportation-related projects based on high-dimensional black-box optimization methods. My current research topics are:

- **RL for LLMs**: Developing an off-policy RL algorithm for post-training of LLMs to solve diverse downstream tasks **([C7], [C13])**.
- **GFlowNets for Scientific Discovery:** Developing an off-policy RL algorithm called GFlowNets to search high-performing and diverse candidates for scientific discovery **([C1], [C2], [C6], [C8], [P2])**.
- **Generative models for Optimization**: Developing a generative model-based approach (Diffusion / Flow-based models) for searching promising candidates in high-dimensional space **([C9], [C10], [C11], [P1])**.

EDUCATION

03/2024 - Current **Ph.D Student in Industrial and Systems Engineering** ***KAIST***

Supervised by Jinkyoo Park

08/2022 - 02/2024 **M.S in Graduate School of AI** ***KAIST***

Supervised by Jinkyoo Park

M.S Thesis: Offline Meta Black-box Optimization Framework for Intelligent Traffic Light Management System

03/2018 - 08/2022 **B.S in Industrial and Systems Engineering & Computer Science** ***KAIST***

INTERNSHIPS

08/2026 - 11/2026 **Research Intern at NaverLabs Europe** ***Grenoble, France***

Hosted by Darko Drakulic

Develop scalable RL algorithm for online multi-agent coordination under uncertainty

06/2025 - 11/2025 **Research Intern at Mila** ***Montreal, Canada***

Hosted by Yoshua Bengio

RL-based red-teaming with evolving environments by safety fine-tuning the victim LLM to promote easy-to-hard exploration.

09/2024 - 03/2025 **Visiting Intern at HKUST** ***Hong Kong (Remote)***

Hosted by Ling Pan

Fine-tuning LLM with GFlowNets to generate diverse and effective prompts for text-to-image diffusion models.

03/2021 - 08/2021 **Research Intern at Kakao Recommendation Team** ***Seoul, Korea***

Develop contextual bandit algorithms for a personal recommendation.

Analyze the gap between simulation and real-world deployment.

INDUSTRIAL PROJECTS

03/2026 - Current **Agentic AI for Traffic-related Question Answering** ***Seoul, Korea***

Collaborate with GS
Develop an Agentic AI system for developing LLM-based for better traffic-related question answering framework.

09/2024 - 09/2025 **Traffic Network Layout Optimization** ***Seoul, Korea***

Collaborate with GS
Develop a Generative model-based design algorithm for optimizing traffic network layout on a given traffic pattern.

03/2023 - 03/2024 **Incentive Design for Managing Taxi Fleet** ***Daejeon, Korea***

Collaborate with ETRI
Develop an RL-based incentive design algorithm for rebalancing taxi fleets to resolve the taxi imbalance problem.

03/2022 - 03/2023 **Traffic Light Optimization** ***Seoul, Korea***

Collaborate with KT
Develop a Bayesian optimization algorithm for managing multiple traffic lights in the real world to reduce congestion.

PUBLICATIONS

*: Equal Contribution

Preprints

Arxiv, 2026 **[P3] Decoupling Exploration and Exploitation in GFlowNets via Policy Evaluation**
Taeyoung Yun*, Inhyuck Song*, Jinkyoo Park, and Ling Pan
Paper / Code

Arxiv, 2026 (based on ICMLW) **[P2] Robust Exploration through Generative Replay**
Inhyuck Song, **Taeyoung Yun**, Jaewoo Lee, and Jinkyoo Park
Paper / Code

Arxiv, 2025 **[P1] Improving Sampling Distribution of Off-policy Training in Generative Flow Networks**
Taeyoung Yun, Sujin Yun, Jinkyoo Park, and Ling Pan
Paper / Code

Conference Publications

NeurIPS, 2026 (based on NIPSW) **[C17] Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization**
Kiyoung Om*, Kyuil Sim*, **Taeyoung Yun***, Hyeongyu Kang, and Jinkyoo Park
Paper / Code

NeurIPS, 2026 (based on ICMLW) **[C16] Aligning Few-Step Generative Models by Amortizing Sample-based Variational Inference**
Jaewoo Lee*, Hyeongyu Kang*, Dohyun Kim, Kyuil Sim, Woocheol Shin, Minsu Kim, **Taeyoung Yun**, Jeongjae Lee, Sanghyeok Choi, Tabitha Edith Lee, Jongchul Ye, Jinkyoo Park
Paper / Code

NeurIPS, 2026 (based on ICMLW) **[C15] Automated Kernel Discovery for Understanding High-dimensional Bayesian Optimization**
Taeyoung Yun*, Woocheol Shin*, Inhyuck Song, Jaewoo Lee, and Jinkyoo Park
Paper / Code

CIKM, 2026 (Short Paper)
[C14] SUMO-SimQA: A Benchmark for LLM Cross-File Reasoning over Traffic Simulation Files
Jeeyun Chang*, **Taeyoung Yun***, Minwoo Jeong, Jinkyoo Park, and Yoonjin Yoon
Paper / Code

ICML, 2026
[C13] Active Attacks: Red-teaming LLMs via Adaptive Environments
Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim
Paper / Code

ICLR, 2026
[C12] Diffusion Alignment as Variational Expectation-Maximization
Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, **Taeyoung Yun**, Kiyoung Om, and Jinkyoo Park
Paper / Code

WSDM, 2026
[C11] Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models
Taeyoung Yun, Inhyuck Song, Woocheol Shin, Yujin Shin, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
Paper / Code

KDD, 2025
[C10] Wind Farm Layout Optimization with Diffusion Models
Yujin Shin*, **Taeyoung Yun***, Sujin Yun, Sungpil Woo, Sunhwan Lim, and Jinkyoo Park
Paper / Code

ICML, 2025 (based on ICLRW)
[C9] Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization
Taeyoung Yun*, Kiyoung Om*, Jaewoo Lee, Sujin Yun, and Jinkyoo Park
Paper / Code

ICML, 2025 (based on NIPSW)
[C8] Improved Off-Policy Reinforcement Learning in Biological Sequence Design
Hyeonah Kim, Minsu Kim, **Taeyoung Yun**, Sanghyeok Choi, Emmanuel Bengio, Alex Hernández Garcia, and Jinkyoo Park
Paper / Code

CVPR, 2025
[C7] Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation
Taeyoung Yun, Dinghuai Zhang, Jinkyoo Park, and Ling Pan
Paper / Code

ICLR, 2025
[C6] Adaptive Teachers for Amortized Samplers
Minsu Kim*, Sanghyeok Choi*, **Taeyoung Yun**, Emmanuel Bengio, Leo Feng, Jarrid Rector-Brooks, Sungsoo Ahn, Jinkyoo Park, Nikolay Malkin, and Yoshua Bengio
Paper / Code

NeurIPS, 2024
[C5] Guided Trajectory Generation with Diffusion Models for Offline Model-based Optimization
Taeyoung Yun, Sujin Yun, Jaewoo Lee, and Jinkyoo Park
Paper / Code

NeurIPS, 2024 (based on ICLRW)
[C4] GTA: Generative Trajetory Augmentation with Guidance for Offline Reinforcement Learning
Jaewoo Lee*, Sujin Yun*, **Taeyoung Yun**, and Jinkyoo Park
Paper / Code

KDD, 2024
[C3] An Offline Meta Black-box Optimization Framework for Adaptive Design of Urban Traffic Light Management Systems
Taeyoung Yun*, Kanghoon Lee*, Sujin Yun, Ilmyung Kim, Won-Woo Jung, Min-Cheol Kwon, Kyujin Choi, Yoohyeon Lee, and Jinkyoo Park
Paper / Code

ICML, 2024 (based on NIPSW)
[C2] Learning to Scale Logits for Temperature-conditional GFlowNets
Minsu Kim*, Juhwan Ko*, **Taeyoung Yun***, Dinghuai Zhang, Ling Pan, Woochang Kim, Jinkyoo Park, and Yoshua Bengio
Paper / Code

ICLR, 2024 (Spotlight)

[C1] Local Search GFlowNets
Minsu Kim, **Taeyoung Yun**, Emmanuel Bengio, Dinghuai Zhang, Yoshua Bengio, Sungsoo Ahn, and Jinkyoo Park
Paper / Code

Journal Publications

TMLR, 2026 (Survey Certification)

[J1] Offline Model-Based Optimization: Comprehensive Review
Minsu Kim, Jiayao Gu, Ye Yuan, **Taeyoung Yun**, Zixuan Liu, Yoshua Bengio, Can Chen
Paper / Code

TEACHING EXPERIENCES

2026 **Teaching Assistant** ***KAIST***
IE579: Game Theory and Multi-Agent Reinforcement Learning

2025 **Teaching Assistant** ***KAIST***
IE343: Statistical Machine Learning
IE437: Data-Driven Decision Making and Control

2024 **Teaching Assistant** ***KAIST***
IE437: Data-Driven Decision Making and Control
IE481: Manufacturing & Artificial Intelligence

2022 **Teaching Assistant** ***KAIST***
MAS480: Introduction to Scientific Machine Learning

ACADEMIC SERVICES

2025 **Reviewer**
ICLR, AAMAS, AISTATS, ICML, KDD, TMLR, NeurIPS

2026 **Reviewer**
AAAI, KDD, ICLR, AAMAS, AISTATS, TMLR, TPAMI, ICML, RLC, NeurIPS

INVITED TALKS

2026 **AutoML Seminar**
Posterior Inference in Generative Models for High-dimensional Black-box Optimization

HONORS & AWARDS

2025 **Qualcomm Innovation Fellowship Finalist** ***Seoul, Korea***
Paper: Learning to Sample Effective and Diverse Prompts for Text-to-Image Generation

2025 **Top Reviewer** ***NeurIPS***
Honor for Top 8% Reviewers

2021 **Dean's List** ***KAIST***
Honor for Top 2% Students

2021 **Excellence Award (2nd Place)** ***Seoul, Korea***
Big Data Competition Hosted by NH